ENGINEERING ASSIGNMENT / NETDRV-001

Linux E1000 Network Driver

Objective: Develop a Linux kernel driver for the Intel 82540EM Gigabit Ethernet Controller emulated by QEMU. The driver must provide reliable Ethernet communication through the Linux networking stack, handle device lifecycle events correctly, and remain stable under sustained network traffic.

Completion rule: The project is complete when all mandatory acceptance criteria pass.

Project scope

Hardware	QEMU Intel 82540EM (8086:100e)
Operating system	Linux, current development kernel
Architecture	x86-64
Build	Loadable kernel module
Network	Ethernet, 1 Gbps controller
MTU	Standard 1500 bytes
CPU configuration	2 vCPUs minimum
Reference	Intel hardware documentation and existing Linux drivers

The test environment must include an independently functioning management network interface and a traffic endpoint reachable through the E1000 device.

1. Functional requirements

NET-001 — Device discovery and initialization

- Driver must bind to the supported PCI device.
- Driver must initialize the device into an operational state.
- Driver must correctly acquire and release hardware resources.
- Driver must report initialization failures without leaving partially initialized devices.

Acceptance criteria

- Device binds automatically when the module is loaded.
- Device initialization completes without kernel warnings.
- Unsupported PCI devices are not claimed.
- Unloading the module releases the device.

NET-002 — Network interface

- Driver must expose a standard Linux Ethernet network interface.
- Interface must have a valid MAC address.
- Interface must support administrative UP and DOWN states.
- Interface must report carrier state accurately.
- Interface must support an MTU of 1500 bytes.

Acceptance criteria

- Interface appears in ip link.

- MAC address matches the configured emulated NIC address.
- Interface can repeatedly transition UP → DOWN → UP.
- Interface reports correct administrative and carrier states.
- No kernel warnings occur during state transitions.

NET-003 — Packet transmission

- Driver must transmit valid Ethernet frames.
- Driver must support standard Ethernet frame sizes.
- Driver must preserve transmitted frame contents.
- Driver must correctly handle TX descriptor exhaustion.
- Driver must reclaim transmission resources after completion.
- Driver must not silently discard packets accepted for transmission because the TX ring is full.

Acceptance criteria

- Frames of varying lengths, from minimum Ethernet size to the standard maximum, are successfully transmitted.
- An independent receiver verifies frame contents.
- Transmission continues after descriptor-ring wraparound.
- Transmission continues after temporary TX queue saturation.
- No permanent TX queue stalls occur during testing.

NET-004 — Packet reception

- Driver must receive valid Ethernet frames.
- Driver must deliver received packets to the Linux networking stack.
- Driver must preserve received frame contents.
- Driver must manage receive buffers correctly.
- Driver must handle RX descriptor-ring wraparound.
- Driver must safely handle invalid lengths and hardware-reported receive errors.

Acceptance criteria

- Valid incoming Ethernet frames reach the networking stack.
- Received data matches transmitted data.
- Frames of varying supported sizes are received successfully.
- Reception continues after descriptor-ring wraparound.
- Invalid packets are not passed to the networking stack as valid frames.

NET-005 — Interrupts and packet processing

- Driver must support hardware-generated interrupts.
- Driver must use the Linux NAPI mechanism for normal packet processing.
- Driver must correctly handle TX and RX completions.
- Driver must not lose work when interrupts and packet processing overlap.
- Driver must not generate uncontrolled interrupt activity.

Acceptance criteria

- TX and RX operate without continuous busy polling.
- Sustained traffic does not cause packet processing to stop.
- Concurrent TX and RX work correctly.
- Interrupt processing remains functional across repeated interface restarts.

NET-006 — Network protocols

The driver is responsible for Ethernet frame transport, not implementing higher-level protocols.

Acceptance criteria

Protocol / traffic	Expected result
ARP	Successful address resolution
IPv4	Successful packet delivery
ICMP	Successful echo request/reply
UDP	Bidirectional payload delivery
TCP	Successful connection and data transfer

These must operate over the E1000 interface, independently of the management NIC.

2. Reliability requirements

REL-001 — Device lifecycle

- Device must initialize correctly after repeated module loading.
- Device must shut down cleanly.
- Interface must support repeated UP/DOWN transitions.
- Device resources must not be accessed after release.
- Driver must clean up correctly after initialization failures.

Acceptance criteria

- 50 consecutive module load/unload cycles without failure.
- 100 consecutive interface UP/DOWN cycles without failure.
- No kernel oops, warning, use-after-free, or resource leak.
- Device remains usable after each cycle.

REL-002 — Sustained operation

- Driver must operate continuously under sustained network traffic.
- TX and RX must work simultaneously.
- Descriptor rings must not become permanently stalled.
- Driver must preserve data integrity under load.

Acceptance criteria

- 30 minutes of sustained bidirectional TCP traffic without a driver failure.
- At least 100,000 frames transmitted and received per direction across controlled tests.
- No TX or RX queue stalls requiring interface restart.
- No application-level payload corruption.
- No unexpected kernel warnings.

REL-003 — Error handling

- Driver must handle resource allocation failures safely.
- Driver must handle packet allocation failures safely.
- Driver must handle unavailable or exhausted descriptors correctly.
- Driver must prevent resource leaks on partial initialization.
- Driver must remain in a valid state after recoverable errors.

Acceptance criteria

- Tested failure paths release all previously acquired resources.
- No double frees or invalid DMA accesses.

- Recoverable errors do not leave the interface permanently unusable.
- Nonrecoverable initialization errors are reported and result in clean teardown.

3. Quality requirements

ID	Requirement	Acceptance criterion
QLT-001	Kernel coding standards	No relevant checkpatch.pl errors; warnings reviewed
QLT-002	Build quality	Builds against the target kernel with W=1 and no new warnings
QLT-003	Memory safety	No detected memory safety violations in applicable sanitizer tests
QLT-004	Concurrency	No lockdep reports or observed concurrency failures during stress tests
QLT-005	DMA correctness	No DMA API debug violations in tested paths
QLT-006	Resource ownership	All allocations, mappings, and registrations have correct cleanup
QLT-007	Maintainability	Code is reviewed, with no unresolved correctness findings

4. Formal acceptance test matrix

All 16 tests are mandatory for project completion. Status is to be completed by the engineer/reviewer.

ID	Test	Expected result	Status
T01	PCI device binding	Driver binds to 8086:100e	_____
T02	Interface registration	Valid Ethernet interface appears	_____
T03	Interface lifecycle	100 successful UP/DOWN cycles	_____
T04	TX frame sizes	Correct transmission across supported frame sizes	_____
T05	RX frame sizes	Correct reception across supported frame sizes	_____
T06	ARP	Address resolution succeeds	_____
T07	ICMP	1,000 echo requests, zero loss on idle controlled link	_____
T08	UDP	Bidirectional payload integrity verified	_____
T09	TCP	Bidirectional data transfer verified	_____
T10	Ring wraparound	100,000+ frames per direction without stalling	_____
T11	TX saturation	Traffic resumes after temporary queue exhaustion	_____
T12	Concurrent traffic	30 minutes of bidirectional traffic without driver failure	_____
T13	Module lifecycle	50 successful load/unload cycles	_____
T14	Failure handling	Injected failures produce correct cleanup	_____
T15	Kernel diagnostics	No unexplained warnings, memory or DMA violations	_____
T16	Management isolation	VirtIO management connection remains functional throughout E1000 tests	_____

A test is passed only when its expected result is observed and recorded. Test setup, traffic rates, packet sizes, and relevant kernel/QEMU versions must be documented so results are reproducible.

5. Explicitly out of scope

- TCP segmentation offloading (TSO).
- Checksum offloading.
- VLAN acceleration.
- Jumbo frames.
- Multiple TX/RX queues.
- Advanced ethtool operations.
- Wake-on-LAN and suspend/resume.
- Physical hardware validation.
- Performance parity with the upstream E1000 driver.

Constraint: The driver must not advertise an optional capability that it does not correctly implement.

6. Required deliverables

1. Driver source code. Integrated with the kernel build system and buildable as a module.

2. Test evidence. Results for all 16 acceptance tests, including relevant logs and reproduction details.

3. Technical documentation. Supported hardware, limitations, configuration assumptions, and known issues.

4. Code review. No outstanding correctness issues classified as blocking.

5. Final demonstration. Working interface, bidirectional communication, sustained traffic, and successful unload/reload.

Definition of done

The assignment is accepted when:

- All mandatory functional requirements are satisfied.
- All 16 acceptance tests pass.
- No known memory safety, DMA, locking, or lifecycle defects remain.
- Source code and testing documentation are delivered and reviewed.

Environment limitation: QEMU user-mode networking supports basic connectivity, but a dedicated Ethernet peer is needed for comprehensive traffic tests. QEMU emulation cannot establish correctness on physical Intel hardware.

Scope boundary: A functionally correct, reliable Ethernet driver for one emulated device, not a feature-complete replacement for the upstream E1000 driver.